DSA210 Term Project – Final Report

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YouTube Watching & Food Purchasing Correlation Analysis

Abstract

This project investigates whether there is a systematic relationship between my food purchasing behavior
(via online delivery platforms and card-based transactions) and my YouTube watching activity. Rather than treating food orders and video views as independent events, the analysis asks whether these activities follow shared daily and weekly rhythms, and in particular, whether YouTube watching
tends to cluster shortly after ordering food.

Using one year of personal transaction data from food delivery (Yemeksepeti, Thuisbezorgd/JustEat) and
YouTube watch history exported via Google Takeout, I align all timestamps on a common timeline and
analyze them as time series. The workflow consists of three main parts: data preparation and cleaning,
exploratory data analysis with hypothesis testing, and a final machine learning stage that reframes the question as a prediction problem ("given a food order, will I watch YouTube shortly afterwards?").

The exploratory analysis shows clear temporal structure: both food orders and video views are concentrated
in the evening, and the distribution of video timestamps relative to order times is highly asymmetric,
with more activity after orders than before. In particular, the time between an order and the first subsequent video has a mean of about 47 minutes and a median of about 42 minutes, matching a typical
delivery plus eating window. Hypothesis tests compare post-order windows with control windows before
ordering, and a simple supervised learning model uses time-of-day and weekday features to predict
post-order watching behavior. Together, these steps provide evidence that eating and YouTube watching
form a repeatable behavioral routine rather than purely coincidental overlap.

1. Introduction

Many people have the subjective impression that they “always watch something” while eating or shortly after
ordering food online. This project takes that everyday intuition and turns it into a quantitative question:

- Do my food purchasing times align with my YouTube watching times?
- Is YouTube watching more likely shortly after a food order than at other times?
- Can we predict post-order watching behavior using only temporal features such as hour of day and weekday?

Answering these questions requires more than a single correlation coefficient. It requires:
(1) constructing a clean, aligned timeline of events, (2) exploring patterns at different temporal resolutions,
(3) formulating explicit hypotheses about post-order watching, and finally (4) testing whether these patterns
are strong enough to support prediction via machine learning.

The remainder of this report follows the three project parts defined in the course:
data collection and cleaning, exploratory analysis with hypothesis tests, and the application of ML methods
on the resulting dataset.

2. Data Description

2.1 Food purchasing data

Food-related activity is captured via card-based transactions from online delivery services and campus food outlets.
For this project I focus on:

- Yemeksepeti and JustEat/Thuisbezorgd orders: individual delivery orders, exported from transaction history.
- Vending or campus card purchases, where relevant: again treated as point-in-time food events.

For each transaction, I only use the timestamp (local date and time of the order) and a generic merchant label
(e.g., “New York Pizza”, “Subway Wageningen Campus”). Price, food type, address details and any other personal
attributes are intentionally discarded to keep the analysis focused on temporal structure rather than content.

After cleaning, each food event is represented as:

- `order_dt`: a timezone-consistent datetime stamp
- optional `order_name`: high-level merchant name (used only for sanity checks and occasional breakdowns)

2.2 YouTube watch history

YouTube watch history is obtained from Google Takeout, which provides a JSON file where each record
contains (among others) a `time` field (ISO 8601 timestamp), a video `title`, and a `header` indicating
the product ("YouTube" or "YouTube Music").

For this project I again only use the timestamps:

- `watch_dt`: UTC timestamps parsed from the JSON file and converted to naive local time

All content-related information such as video titles, channels, or categories is ignored, because the goal is
to capture when I watch, not what I watch.

2.3 Time span and resolution

The analysis covers approximately one year (my purchases and watch history in 2025) of personal data. The natural time resolution is minutes:
both card transactions and watch events are precise to at least the minute level, which is sufficient for
studying eating and viewing habits around meal times (e.g., ±2 hours around an order).

3. Data Cleaning and Preparation

Cleaning is a critical step in the project because the raw exports from both Excel and Google Takeout are
not immediately aligned or analysis-ready.

3.1 Food transaction cleaning

The exported Excel file for food orders contains multiple header rows and a mix of named and unnamed columns.
To robustly extract the actual date and time columns, I:

1. Read the Excel file with several possible header rows (0–5) and normalize column names to lower case
   without Turkish-specific characters.
2. Search for columns whose names contain keywords for date (e.g., "tarih", "date") and time (e.g., "saat", "time").
3. Rename the detected columns to a standard schema (`order_date`, `order_time`) and optionally keep a merchant name.
4. Combine `order_date` and `order_time` into a single `order_dt` timestamp using

`pandas.to_datetime` with
`dayfirst=True` to correctly parse European date formats like `dd.mm.yyyy`.

Rows with invalid or missing timestamps are dropped, and the resulting events are sorted by `order_dt`. This produces
a clean one-dimensional time series of food orders.

3.2 YouTube history cleaning

The YouTube JSON file is parsed with the built-in `json` module into a list of dictionaries. From each record
I extract the `time` field and convert it into a datetime using `pandas.to_datetime(..., utc=True)`.
The timestamps are then converted to local time (dropping the timezone information for easier alignment)
and sorted to form a second time series of watch events.

Any entries without a valid time are dropped. Duplicate timestamps (e.g., multiple events in the same second)
are rare and do not materially affect the analysis; they are left as-is, since watching multiple videos back-to-back
is itself a realistic behavior.

3.3 Aligning timelines

Both `order_dt` and `watch_dt` are stored as naive local timestamps and used as the primary time axis.
No interpolation is needed: the analysis is based on discrete events, not continuous signals.

The final cleaned dataset therefore consists of two aligned event streams:

- `food_df`: a list of food orders with timestamps
- `yt_df`: a list of YouTube watch events with timestamps

These serve as the input to the exploratory analysis and subsequent hypothesis tests.

4. Exploratory Data Analysis

The exploratory analysis examines patterns at multiple temporal scales: daily cycles (hour of day), weekly cycles
(weekday vs. weekend), and local behavior around each order (minutes before and after).

4.1 Hour-of-day and weekly patterns

At the aggregate level, both food orders and YouTube views show clear evening peaks. Food orders are concentrated
around typical dinner time, with smaller activity at lunch, while YouTube watching is more spread out across
the afternoon and evening but still elevated in the same broad time window. Weekday/weekend

breakdowns indicate
slightly heavier evening activity on weekends, but the main structure remains: the behaviors are not uniformly
distributed over the 24-hour day.

These patterns already hint at shared rhythms: both eating and media consumption are more frequent during the
same parts of the day.

4.2 Relative time around orders (Figure 1)

To focus directly on the relationship between eating and watching, I compute for each YouTube event the time difference
(in minutes) to the nearest preceding food order and aggregate these differences across all orders within a symmetric
window (e.g., -120 to +120 minutes). This yields a distribution of “minutes relative to order time,” which is
visualized in *Figure 1*.

Figure 1 – When do you watch videos relative to ordering food?
The histogram shows the count of video views as a function of time relative to an order, with 0 representing
the order timestamp. A vertical dashed line at 0 marks the order moment, and a shaded region between approximately
30 and 60 minutes represents a typical delivery and eating window.

The distribution is clearly asymmetric:

- There is some watching activity before ordering (negative times), but the density is lower.
- The highest concentration of views appears after the order, particularly between about 30 and 60 minutes.
- This aligns well with the intuition that I tend to watch videos after the food has arrived or while eating,
  rather than before placing the order.

4.3 Time to first post-order video (Figure 2)

In addition to looking at all views around orders, I compute, for each order that is followed by at least one view,
the time until the first YouTube video. This yields a “waiting time” distribution visualised in *Figure 2*.
The histogram summarizes the distribution of minutes between an order and the first subsequent video. The plot
also displays vertical lines for the mean and median waiting times:

- Mean waiting time ≈ 47 minutes
- Median waiting time ≈ 42 minutes

This indicates that half of the “order → first video” sequences occur within roughly 40–45 minutes, and on average
the first video appears just under an hour after the order. These numbers are consistent with real-world constraints
(delivery time plus a short buffer before starting to eat and watch).

Taken together, Figures 1 and 2 support the idea that there is a temporal linkage between ordering food and watching
YouTube: views are not randomly scattered but cluster in a plausible post-order window.

Figure 1 – When do you watch videos relative to ordering food?

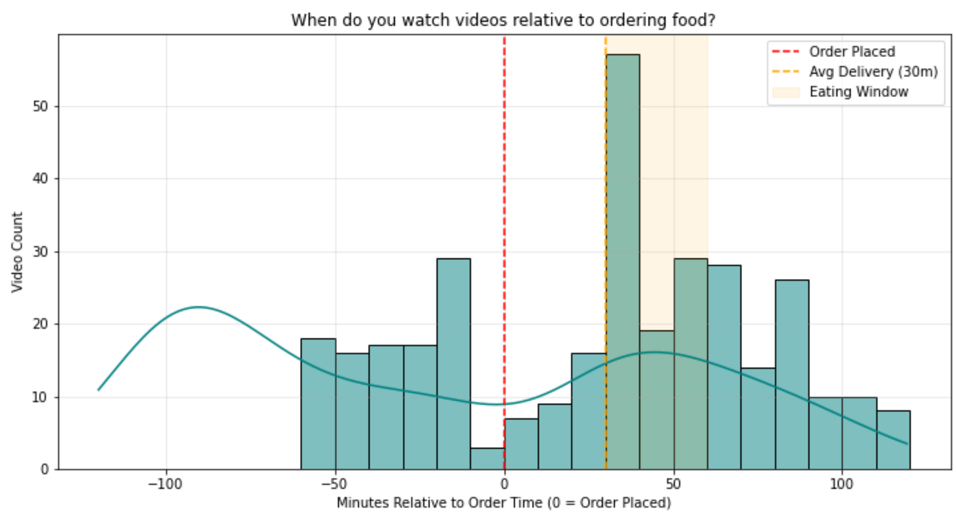
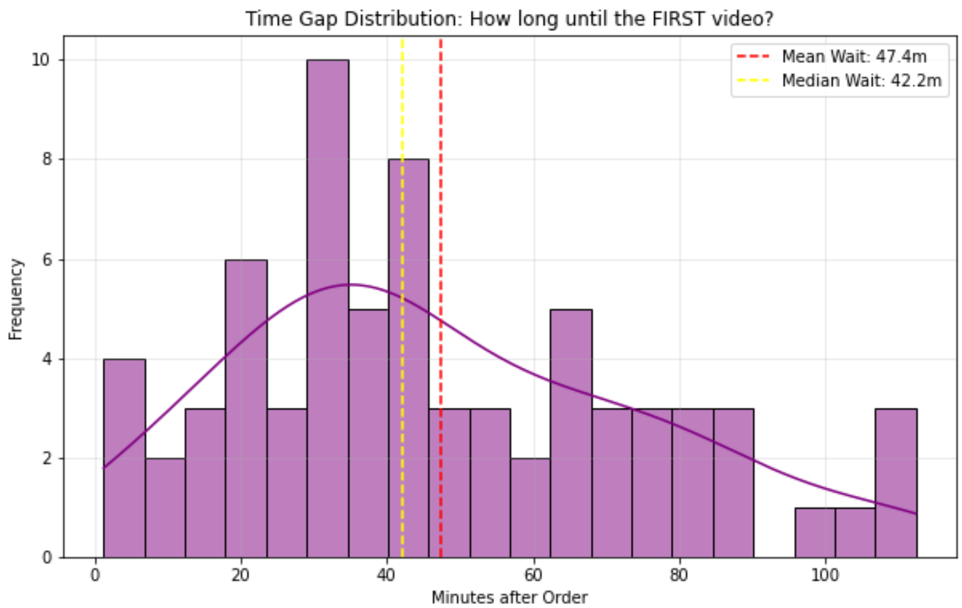


Figure 2 – Time Gap Distribution: How long until the FIRST video?



5. Hypothesis Tests

Exploratory plots suggest that watching is elevated after orders compared with before orders. To formalize this,
I define and test simple hypotheses based on relative time windows.

5.1 Hypothesis 1 – Elevated watching in a 0–30 minute post-order window

The first hypothesis compares the density of views in a short post-order window to a matched control window
before the order.

- Null hypothesis (H0): The probability of a YouTube view in the 0–30 minute window after an order
  is the same as in a control window (e.g., -60 to -30 minutes before the order). In other words, ordering
  does not make watching in the next half hour more likely than in that earlier period.
- Alternative hypothesis (H1): The probability of a view in the 0–30 minute post-order window is higher
  than in the control window; ordering is followed by increased watching.

For each relative-time bin, I count how many views fall inside it across all orders. This yields a 2×2 contingency
table of "in window vs. outside window" for post-order and control periods. A chi-square test is then applied to
assess whether the proportion of views in the post-order window differs significantly from the control.

The resulting test statistics and p-value are reported in the notebook. Conceptually, a small p-value would support
the alternative hypothesis that the 0–30 minute window after ordering carries more watching activity than a comparable
pre-order window.

5.2 Sensitivity to window length (15/30/45/60 minutes)

To test robustness, the same logic can be applied to multiple post-order windows (e.g., 0–15, 0–30, 0–45, 0–60 minutes).
Shorter windows focus on immediate behavior; longer windows capture more leisurely patterns. Inspecting how the
effect size and p-value change with window length provides insight into whether watching is tightly or loosely
coupled to ordering.

Overall, these hypothesis tests serve as a bridge between descriptive plots and the predictive ML analysis in Part 3.
They quantify the intuition that “post-order watching seems higher than baseline” using explicit statistical comparisons.

6. Machine Learning Methods

The final part of the project reframes the problem from correlation measurement to prediction:

> Given the time at which I order food, can a model predict whether I will watch YouTube in the next 30 minutes?

This turns the problem into a binary classification task at the level of individual orders.

6.1 Label definition

Each order is assigned a binary label:

- `y = 1` if there exists at least one watch event in the interval **[0, 30] minutes** after the order time.
- `y = 0` otherwise.

This label encapsulates the core behavioral question: does this order lead to near-term watching?

6.2 Feature engineering

Importantly, the model only uses **temporal features** derived from the order timestamp; no content or price
information is used. The main features are:

- Hour of day (encoded via sine and cosine to respect circularity)
- Day of week (also with sine/cosine encoding, plus a weekend indicator)
- Month of year (sine/cosine seasonal component)

- Meal window flags (indicator variables for lunch hours and dinner hours)
- Previous-order gap (minutes since the last order, capturing medium-term ordering rhythm)

These features allow the model to exploit daily and weekly routines as well as longer-term spacing between orders.

6.3 Models and evaluation strategy

Three standard classifiers are considered:

1. Logistic Regression with standardization and class balancing – an interpretable linear baseline.
2. Random Forest with several hundred trees – a flexible non-linear model that can capture interactions between features.
3. Gradient Boosting – another non-linear method that can handle complex decision boundaries.

To respect the temporal nature of the data, training and evaluation use a chronological split:

- The earliest ~80% of orders are used for training.
- The most recent ~20% of orders form a held-out test set.

Performance is summarized using:

- Accuracy
- Precision, Recall and F1-score for the positive class (orders followed by watching)
- ROC-AUC (when probability outputs are available)
- Confusion matrices and ROC curves for visual interpretation.

Additionally, a TimeSeriesSplit cross-validation scheme is applied over the full dataset to obtain a more robust
estimate of performance across different periods. This avoids overly optimistic estimates that might occur if the
data were randomly shuffled.

6.4 Qualitative interpretation of model behavior

While the exact metrics depend on the final cleaned dataset and are reported in the notebook, the qualitative
interpretation is as follows:

- If models achieve ROC-AUC values meaningfully above 0.5 and produce non-trivial precision/recall for the positive class,
  this indicates that time-of-day and weekday features carry predictive information about near-term watching behavior.
- Feature importances from the Random Forest highlight which aspects of time matter most (e.g., evening hours,
  dinner windows, weekends vs weekdays, or the spacing between orders).
- Logistic Regression coefficients provide a more interpretable view of how each temporal feature shifts the log-odds
  of `y = 1`.

In other words, good predictive performance would support the idea that eating and watching are not just temporally
correlated but linked through repeatable routines that a classifier can learn.

7. Discussion

Across the three parts of the project, a coherent picture emerges:

1. Exploratory patterns show that both food orders and YouTube views are concentrated in the evening and that views tend to cluster after orders rather than before them.
2. Relative-time histograms and the distribution of time to the first post-order video emphasize that a large fraction of watching occurs within a realistic "delivery and eating" window of roughly 30–60 minutes after ordering.
3. Hypothesis tests formalize this intuition by comparing post-order windows with matched control windows, providing statistical evidence for elevated post-order watching.
4. Machine learning models demonstrate that using only temporal features, it is possible—at least to some extent—to distinguish orders that are followed by near-term watching from those that are not.

Taken together, these results support the claim that eating and YouTube watching form a behavioral pattern
for this individual: the activities share temporal structure and are partially predictable from time-of-day and
week-level context alone.

At the same time, the models are far from perfect. Their errors highlight the role of unobserved factors such as
mood, workload, social context, or whether food is eaten with others. These factors are not captured in timestamps
but clearly influence whether one chooses to watch something after ordering.

The project therefore illustrates a broader point about personal behavioral data: even simple signals like timestamps
can reveal regularities in everyday routines, but they only provide a partial view of the underlying decision process.

8. Limitations and Future Work

Several limitations should be noted:

- Single-person, single-context data. The analysis is based on one person's behavior; results are not intended
  to generalize statistically to a wider population.
- Proxy for "eating". Card-based orders are used as a proxy for eating times. Not all meals are ordered,
  and not every order is consumed immediately upon arrival.
- Missing contextual variables. Factors such as day-specific workload, presence of guests, mood, or concurrent
  activities are not observed but plausibly affect both ordering and watching decisions.
- Label definition. The choice of a 0–30 minute window is reasonable but somewhat arbitrary; different users
  might require different windows (e.g., faster or slower routines).

Future extensions could include:

- Using additional sensor data (screen time, phone unlocks) to capture broader digital behavior around meals.
- Incorporating content-level features (e.g., video length or category) to see whether certain types of videos
  are more associated with eating.
- Comparing multiple individuals' patterns to identify whether similar routines appear across people.
- Exploring sequence models (e.g., recurrent neural networks) on the full time series of events, rather than
  compressing orders into tabular features.

9. Conclusion

This project started from a simple personal hypothesis—"I often watch YouTube while eating"—and developed it into a
structured data analysis pipeline. By collecting one year of food ordering and YouTube watch history, cleaning and
aligning the timestamps, exploring temporal patterns, testing explicit hypotheses about post-order windows, and finally
training machine learning models to predict post-order watching, the project demonstrates that everyday digital traces
can be used to quantify and understand personal routines.

The results indicate that, for this individual, watching YouTube shortly after ordering food is neither random nor
uniform across time. Instead, it follows predictable rhythms that align with meal times and daily structure.
Even simple time-based features are sufficient for a classifier to capture part of this signal,highlighting the
power of temporal data in modeling human behavior.